

Nikitha Padmanabha

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SUMMARY

Data Scientist specializing in production-grade GenAI systems, RAG pipelines, and LLM applications using Python, LangGraph, FastAPI, and transformer architectures. Experienced in deploying scalable AI workflows with evaluation, observability, and latency optimization for enterprise applications.

EXPERIENCE

Data Scientist

Sept. 2025 – Present

Incedo

Bengaluru, India

- Developed a production-grade GenAI agentic chatbot using LangGraph and LangChain — supporting 500+ internal users across 1,000+ monthly queries — by designing structured agent pipelines and ReAct-style reasoning.
- Built LLM evaluation and monitoring pipeline to compare model results with curated ground truth and extract and analyze model metrics including latency, token usage, throughput, and response quality, enabling data-driven improvements to prompt and pipeline design.
- Improved enterprise chatbot response latency by 85% (20s → 3s) by implementing optimized agent routing for simpler user queries in production.
- Improved recommendation relevance and reduced manual analysis effort by 40% by building state-wise LightGBM models on large-scale datasets and applying Hexbin clustering to analyze behavioral and ticketing patterns at scale.
- Mentored and supervised 3 GenAI interns, guiding them in LLM experimentation, evaluation frameworks, and production deployment practices, ensuring successful completion of 3 AI prototype projects.

Software Engineer

Jan. 2025 – Mar. 2025

KGS Technology

Dallas, TX

- Developed robust ML pipelines using Python, scikit-learn,, increasing execution efficiency by 25%.
- Integrated CI/CD practices with Git to automate model deployment and testing across environments.
- Benchmarked pipeline modules for compute efficiency to optimize resource allocation and reduce latency.

Machine Learning Researcher

Jan. 2024 – Dec. 2024

Physics Dept. UTA

Arlington, TX

- Researched extensively on transformer and attention layer and the working of it to use with weather data.
- Built Transformer based architectures in Python to improve space weather prediction accuracy by 25%
- Benchmarked model latency and throughput across GPU configurations, achieving a 20% performance gain through precision tuning and batch-size optimization

PROJECTS

Bibliomind: Semantic search and Q&A engine over a personal library of 50+ books, retrieving source-attributed answers with chunk-level relevance scores across 200K+ embedded tokens.

TEC Map Forecasting: U-Net with Space Weather Data Implemented a modified U-Net for multi-horizon Total Electron Content forecasting, fusing auxiliary space weather variables; evaluated against quiet vs. storm-time periods with structured train/val/test splits.

3D Image Classification using 4-layer Convolutional Neural Network (CNN) in Python to distinguish 3D brain MRI images as Alzheimer's patients or healthy individuals, achieving a 95% accuracy with the NIFTI dataset.

EDUCATION

University of Texas at Arlington

Aug. 2021 – Dec. 2023

Masters of Science in Computer Science

Arlington, TX

PES University

Aug. 2017 – May 2021

Bachelors of Tech, Major in Electrical Engineering, Minor in Computer Science

Bengaluru, India

SKILLS

Languages: Python, SQL, C, C++, HTML

AI/ML: Transformers, LLMs, RAG, CNNs, PyTorch, Scikit-learn, LangChain, LangGraph, Pandas, NumPy

Deployment: FastAPI, Docker, Model Monitoring, Pipeline Orchestration

Tools: Git, Streamlit, Linux, Jupyter Notebook